

Early Warning Systems for Cryptocurrency Markets: Predicting 'Zombie' Assets Using Machine Learning

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Abstract

The cryptocurrency market harbours a hidden risk: assets that silently disappear from trading, leaving investors stranded. These 'zombie' cryptocurrencies technically exist but become temporarily untradable on exchanges, ranging from weeks to permanent delisting. This study predicts which cryptocurrencies are at risk of becoming zombies using predictors derived from return statistics, trading volume, market capitalization, and asset-specific features. Our sample includes cryptocurrencies listed for at least 210 days between January 2015 and December 2022. We employ various machine learning algorithms and novel explainable AI (XAI) tools, including permutation-based feature importance and partial dependence plots (PDPs), to identify and analyze key factors influencing zombification risk. Our machine learning models achieve 84% out-of-time balanced accuracy in predicting whether a cryptocurrency will become a zombie within next 28 days. Tree-based approaches, particularly random forests, significantly outperform traditional econometric methods. Trading volumes and past returns emerge as the most influential predictors.

Keywords: cryptocurrency, coin, token, classification, prediction, machine learning;

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Abstract

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JEL codes: G11, G15, G17;

1 Introduction

Cryptocurrencies are rapidly evolving over time, becoming increasingly prominent in global financial markets ([Ahmed et al., 2024](#); [Fan Fang and Wu, 2021](#)). While cryptos attract significant interest from both investors and academics ([Almeida and Gonçalves, 2023](#)), most studies focus on the best-performing cryptocurrencies – the market leaders ([Beneki et al., 2019](#); [Będowska-Sójka and Kliber, 2021](#); [Gaies et al., 2024](#); [Yousaf et al., 2024](#)). Far less attention is paid to those who struggle or disappear from the market ([Galati et al., 2024](#); [Martins, 2024](#)). In this study, we examine cryptocurrencies that become temporarily untradable, a phenomenon first highlighted in financial media ([Kharif, 2022](#)) as ‘crypto zombies.’ We expand this concept to include all cryptocurrencies that lose their tradability, regardless of their listing status or potential for market recovery.

We aim to identify key predictors signalling a cryptocurrency’s transition to this untradable state. We search for those predictors among three groups. The first one is based on descriptive statistics of prices, volumes, and market capitalization which are easily available measures. These variables are deeply embedded in behavioural finance, particularly in concepts such as herd behaviour ([Gemayel and Preda, 2024](#); [Papadamou et al., 2021](#); [Vidal-Tomás et al., 2019](#)), loss aversion ([Li et al., 2021](#)), and investor sentiment ([Angeles López-Cabarcos et al., 2021](#)). The second group consists of more advanced features that require longer time series, such as the long memory parameter ([Bariviera et al., 2017](#)), estimates of an alpha-stable tail index ([Pele et al., 2021](#)), Value-at-Risk, and Expected Shortfall ([Harasheh and Bouteska, 2025](#); [Jiménez et al., 2022](#)). Finally, following [Groby and Sapkota \(2020\)](#), we include categorical predictors capturing asset-specific features.

Numerous studies utilize machine learning to predict cryptocurrency prices ([El-Khatib and Hatemi-J, 2022](#); [Oyedele et al., 2022](#); [Ren et al., 2022](#); [Wang and Chong, 2021](#)). [Cakici et al. \(2024\)](#) apply the machine learning methods to predict the cross-section of cryptocurrency returns. They discuss ‘factor-zoo’ which potentially impacts returns of crypto-assets and point out that past alpha, illiquidity, and momentum are the prevailing factors which allow improving forecasts. In a different vein, here we focus on the classification exercise predicting ‘dead or alive’ assets utilizing some of the features described in the literature ([Cakici et al., 2024](#); [Pele et al., 2021](#)) such as alpha stability, volatility or volume of the cryptocurrency. In a similar vein, [Wang et al. \(2023\)](#) apply machine learning models to predict volatility of cryptocurrencies. They find that internal determinants, such as previous trading information or lagged volatility, play major roles when forecasting volatility.

The literature examines various factors affecting cryptocurrency prices. [Demir et al. \(2018\)](#) find that the Economic Policy Uncertainty index improves predictions of bitcoin returns. [Galati et al. \(2024\)](#) document spillover effects between cryptocurrency exchanges after the FTX collapse, showing investors shift from minor to established cryptocurrencies in a flight-to-safety pattern. However, these studies don’t address factors that lead cryptocurrencies to become untradable. The most closely related paper to ours is [Milunovich and Lee \(2022\)](#), which compared machine learning models for predicting which of the cryptocurrency exchanges remain active. Our approach differs from previous research by focusing on predicting when individual cryptocurrencies may become untradable – whether temporarily or permanently.

We test two research hypotheses. The first is that an investor can predict whether a cryptocurrency will become a zombie with relatively high balanced accuracy. The second hypothesis is that non-linear machine learning algorithms outperform the traditional parametric model due to the potentially complex, non-linear relationships between the

risk of becoming a zombie and its predictors. We find that the random forest followed by xgboost provides the highest balanced accuracy in detecting zombies and non-zombies. The application of various non-linear machine learning algorithms and data rebalancing techniques allows us to achieve relatively high out-of-sample performance. In our test sample, we can predict that crypto will become a zombie within the next 28 days with 84% balanced accuracy. We find that the variables with the greatest explanatory power include minimum and maximum volumes, and median returns calculated based on previous periods. In some specifications also a category of the cryptocurrency, its average or median market capitalisation as well as the first and third quantile of returns distribution are important predictors of becoming a zombie.

We contribute to the existing literature in several ways. First, while previous studies have developed models to predict cryptocurrency bankruptcy (Fantazzini, 2023; Feder et al., 2018; Grobys and Sapkota, 2020; Ma et al., 2022), our research broadens this scope to include cryptocurrencies we classify as zombies they experience no trading for up to four weeks without being officially delisted. This expanded focus captures both permanent and interim inactivity, offering a more holistic view of risks specific to cryptocurrencies. Second, we enrich the approach of Ma et al. (2022), Gu et al. (2020) and Milunovich and Lee (2022) by performing a comparative analysis of machine learning methods. Similarly to Grobys and Sapkota (2020) we focus on the prediction of the emergence of zombies based on historical data on prices and volumes, but also extend the range of variables including less obvious measures. Contrary to their work, we show that distinguishing between proof-of-work and proof-of-stake does not add any value in predicting which crypto will become a zombie. Moreover, we show that adding more features to the model has limited benefit in predicting becoming zombies. Third, to make our machine learning models interpretable (Tang et al., 2024), we employ eXplainable Artificial Intelligence (XAI) tools to analyze these black box algorithms. Thus we can indicate the most important predictors which are behind becoming a zombie. Our results show that the most important are variables based on volume. Fourth, we contribute to the literature by analysing a relatively long period from the emergence of cryptocurrencies until the end of 2022 and consider all cryptocurrencies traded within this period thus avoiding the survivorship bias.

To achieve this, we constructed a database consisting of weekly snapshots of nearly 9.5K cryptocurrencies listed in the period from January 2015 to December 2022. We then perform the empirical exercise, in which we apply several machine learning techniques such as logistic regression, LASSO, random forests and extreme gradient boosting to predict the 'zombie'-state of the crypto within the next 28 days. A variety of performance measures are used to indicate the best model in terms of prediction accuracy.

The detection of potential zombies is important for several reasons. First, it helps to protect investors from misguided investments. Second, avoiding those cryptocurrencies that may disappear from the market or become non-tradable for a given period is critical for risk management. Third, knowing which cryptocurrencies are at risk of becoming zombies would be beneficial for anyone building a portfolio of cryptocurrencies. Tools to identify such risk are important, especially in a rapidly evolving market with constantly changing regulations¹.

¹From a supervisory point of view, the regulators deal mainly with exchanges. In 2022 in the US, a new directive gave the power to regulators of the financial market to control cryptocurrency exchanges. More recently, in March 2023 the development of Venezuela's cryptocurrency market was significantly reduced (<https://decrypt.co/124525/shut-down-mining-farms-exchanges-venezuela>, access on April 25, 2023). Chinese authorities entered regulatory change and almost banned cryptocurrencies in May 2021 (<https://www.loc.gov/item/global-legal-monitor/2021-10-13/china-central-bank-issues-new-regulatory-document-on-cryptocurrency-trading/>, access on April 25, 2023). This resulted in the huge drop in trading volume observed in the Chinese cryp-

2 Methodology

In our analysis, we use classification models to estimate the probability that a particular crypto-asset will become a zombie during the next k days. To better capture the zombification process, we use $k \in \{7, 28\}$, i.e. we perform the prediction for the next week or the next four weeks. Let T denote the number of temporal observations recorded and t_0 the first time point used for analysis. Then at the end of every Sunday in our sample ($t \in [t_0, T]$), we define the binary variable $Y_{i,t}^k$ for crypto-asset i as follows:

$$Y_{i,t}^k = \begin{cases} 1 & \text{if crypto-asset } i \text{ becomes "zombie" within the next } k \text{ days from time } t \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The classification model f estimates the probability $\hat{P}(Y_{i,t}^k = 1 | \mathbf{X}_{i,t})$ of a cryptocurrency becoming a zombie:

$$\hat{P}(Y_{i,t}^k = 1 | \mathbf{X}_{i,t}) = f(\mathbf{X}_{i,t}), \quad (2)$$

where $\mathbf{X}_{i,t}$ represents the features matrix for cryptocurrency i at time t (the features are detailed in Section 3).

The models are estimated and evaluated using cross-validation on a rolling basis with 7 iterations. The whole dataset is divided into training and testing samples based on time.

Each time, we use a rolling 3-year period to train the model, and we check its performance on the out-of-time validation sample covering the next 6 months, as illustrated in Figure 1. Averaged results on the seven out-of-time validation periods (2019H1 – 2022H1) are used to tune hyperparameters with the grid search and select the best modelling approach (Hastie et al., 2009). The final training sample from 2019H2 to 2022H1 (dark green background on the plot) was used to re-estimate the best model on the newer data and verify its performance on the final test sample (2022H2).

We implement a range of widely adopted machine learning algorithms (Almeida and Gonçalves, 2023; Dudek et al., 2024; Nagula and Alexakis, 2022) and compare them using various performance metrics. First, we apply the standard logistic regression (henceforth GLM), which serves as a simple benchmark. We also use the least absolute shrinkage and selection operator (LASSO) (Tibshirani, 1996) which is an extension of the GLM, that penalizes features that have low predictive power by shrinking their coefficients towards zero. Some of the coefficients might be even set to zero, which is why LASSO applies variable selection. We also apply two tree-based models, such as random forests (RF) and extreme gradient boosting (XGB). Random forest (Breiman, 2001) combines the predictions of multiple independent decision trees, while extreme gradient boosting (Chen and Guestrin, 2016) combines the predictions of multiple decision trees estimated sequentially.

Comparing various ML approaches in fraud detection (Achakzai and Juan, 2022) or mispricing identification in financial markets (Hanauer et al., 2022) is a common practice. For each approach, a variety of classification performance measures is applied. The area under the Receiver Operating Characteristic curve (AUROC) is a single metric indicating the performance of a model independently of the selected cut-off point. We also take into account Balanced Accuracy which is an arithmetic average of Sensitivity and Specificity and better than simple accuracy reflects the predictive performance of models in case

of cryptocurrency market and Bitcoin market globally (Borri and Shakhnov, 2020). Canadian regulators were among the most proactive, allowing for the launching of bitcoin exchange-traded funds in 2021 (<https://evolveetfs.com/2021/02/evolve-announces-bitcoin-etf-begins-trading-on-tsx/>).

Figure 1: Model validation scheme

	2016H1	2016H2	2017H1	2017H2	2018H1	2018H2	2019H1	2019H2	2020H1	2020H2	2021H1	2021H2	2022H1	2022H2
validation 1														
validation 2														
validation 3														
validation 4														
validation 5														
validation 6														
validation 7														
test														

Note: Green background indicates 3-yearly training samples, orange – 6-monthly validation samples used to tune hyperparameters and select the best model, red – final test sample.

of highly imbalanced data (Yan et al., 2022). In the context of our study Specificity counts the share of correctly predicted non-zombies among all real tradable coins, while Sensitivity counts the share of correctly predicted zombies among all real zombies. This metric is crucial because misclassifying a zombie as an active cryptocurrency carries higher potential costs for investors than the opposite error.

Last, but not least, we apply explainable artificial intelligence (XAI) tools to identify which features are the most important predictors of being zombie and understand their impact on the outcome variable. We use model-agnostic methods that are easily comparable independently of the structure of a model. First, the permutation-based feature importance metric is used to identify important predictors. It calculates how the selected model performance measure changes if the values in a particular feature are randomly permuted. Second, we use partial dependence profiles that show how the expected model prediction changes with respect to the values of the most important explanatory variable keeping all other predictors constant (Molnar, 2022).

3 Data

Our sample is based on all crypto assets listed on CoinMarketCap, available at www.coinmarketcap.com from January 1, 2013, to December 31, 2022. CoinMarketCap provides data on cryptocurrency prices, volumes, market capitalization and circulating supply. It utilizes different algorithms to filter out outliers and avoid inaccurate data (CoinMarketCap, 2023). This database combines information from hundreds of various crypto exchanges to gather a wide range of data. Despite some comments on the veracity of the information contained (Alexander and Dakos, 2020), it continues to be the most widely used database in the research community (Anghel, 2021; Burggraf, 2021).

Out of all the cryptoassets available there in the sample period, we chose those that were listed uninterruptedly for at least 210 days. We require sufficiently long time series to obtain reliable estimates of higher-order statistical measures, including kurtosis, skewness, and the parameters of the α -stable distribution. Crypto assets that maintain continuous market quotation coverage on CoinMarketCap are classified here as 'active'. Based on the available data, we were able to indicate when particular cryptos were traded and when

they stopped being covered and became 'inactive'. If the loss in coverage lasted for at least 28 days they were classified as zombies.

Apart from historical quotations, CoinMarketCap also provides some metadata on each symbol, including the date of their first and last quotation. This information was used to calculate the number of added (or new) cryptocurrencies, actively traded cryptos as well as those which become untradable for each day of our sample.

Figure 2 shows the cumulative number of cryptocurrencies that gained coverage on CoinMarketCap and those that lost it, which made them zombies. We also show a number of currently traded cryptos obtained as a cumulative difference between the former and the latter. These assets exhibit regular price movements and substantial trading volumes, indicating consistent market participation. In contrast, inactive crypto-assets experience periods without market quotation coverage, typically characterized by extended intervals of zero trading volume and static prices. The total number of cryptocurrencies has increased exponentially over time, whereas the number of zombie cryptocurrencies, after an initial rise, has stabilized since 2020. Active crypto-assets predominate in the market, reflecting the ongoing development and maturation of the cryptocurrency ecosystem. Zombie cryptocurrencies which have lost market quotation coverage represent a smaller proportion of the total sample, confirming the developing status of the market. This imbalance between the number of active cryptos and zombies requires applying the correction mechanisms in the empirical part.

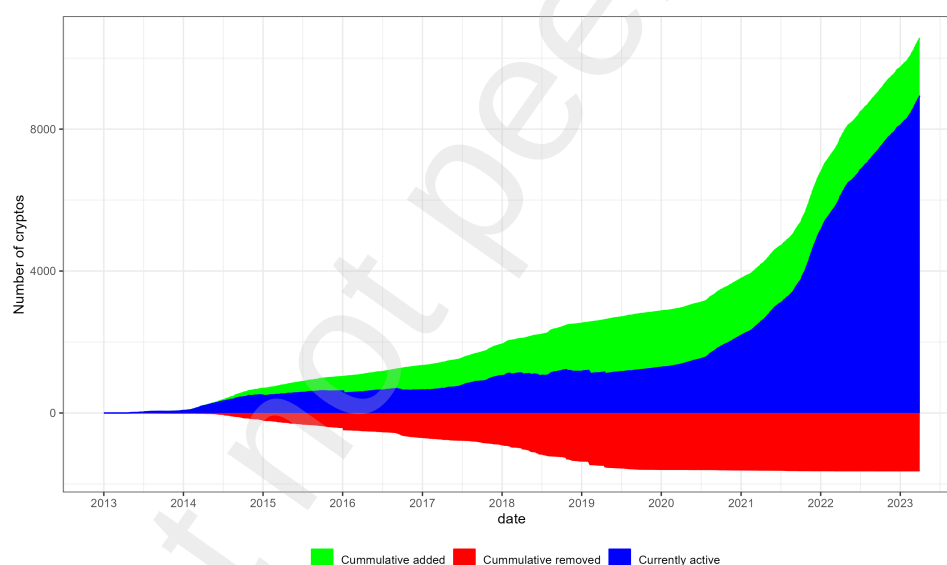


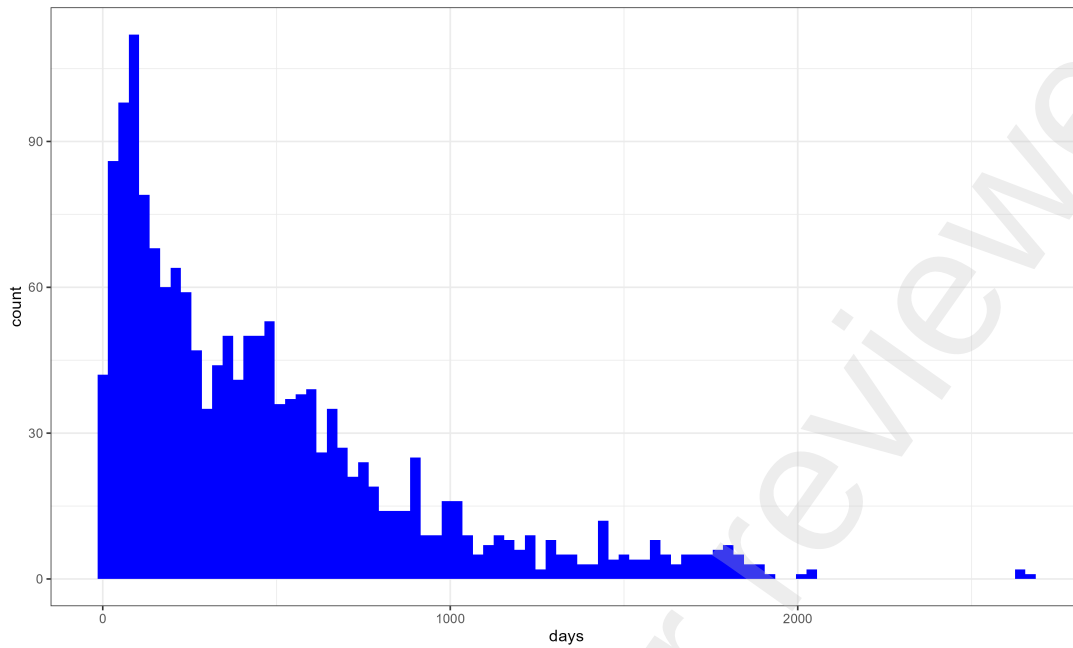
Figure 2: The number of cryptocurrencies: appearing for the first time, zombies, and actively traded over the sample period

Note: The figure shows the number of cryptocurrencies in our dataset that were uninterruptedly quoted for at least 210 days between January 4, 2015, and December 31, 2022, on coinmarketcap.com. Cryptocurrencies that gained market quotation coverage for the first time are presented in green. Zombies that lost market quotation coverage for at least 28 consecutive days are in red. The cumulative value is calculated as the sum of both groups of crypto assets.

The distribution of the life length of such crypto assets is presented in Figure 3. It shows that the possibility of losing market quotation coverage decreases the longer it maintains price tracking in the market. This argues for the inclusion of a variable measuring the length of quotations in the model.

We have created a database of weekly snapshots comprising all cryptocurrencies with available historical data, including those that subsequently lost market quotation cover-

Figure 3: The distribution of life length of zombies



Note: The x-axis shows the number of listing days, while the y-axis shows the number of crypto assets with that length of listing.

age. For every Sunday between 2015-01-04 and 2022-12-25 and every cryptocurrency, we obtained a set of measures based on historical returns, trading volumes, and market capitalisation from the preceding 28 days (4 weeks) and 182 days (26 weeks, ca. half a year). The set of measures consists of statistics including mean, median, minimum, maximum, standard deviation, skewness, kurtosis, first quartile of the distribution (Q1) and third quartile (Q3).

In addition, based only on returns and only on the previous 182 days we calculated the Hurst exponent, α -stable tail index, expected shortfall (ES) and value-at-risk (VaR). The Hurst exponent, which captures long-range dependence, has been observed to vary with the stage of market development. In Bitcoin's early years, the Hurst exponent was high, indicating strong long-range dependence. Over time, it has stabilized and diminished, suggesting the market has transitioned to a more developed stage (Bariviera et al., 2017). The α -stable distributions, commonly used to model heavy-tailed and asymmetric data, can provide a robust framework for analyzing cryptocurrency market behaviour (Raubitzek and Neubauer, 2021). Furthermore, ES and VaR are crucial risk measures that can help evaluate potential losses beyond the expected risk level (Dutta et al., 2020). The variables described above are used in the basic specification of the model (S1).

Next, we expand a list of predictors by four fixed categorical predictors used in similar studies (Fantazzini, 2023; Grobys and Sapkota, 2020; Gu et al., 2020; Ma et al., 2022). Specification 2 (S2) in addition to S1 includes the category of the crypto-asset, namely coin versus token, two dummy variables indicating whether a symbol has a Twitter (X) and separate subreddit account, and the age of the cryptocurrency in days (since its first quotation). The widest Specification 3 (S3) includes three additional dummy variables based on tags provided by CoinMarketCap indicating if a symbol is mineable, proof-of-stake (pos) or proof-of-work (pow).

The list of all characteristics of cryptocurrencies used as predictors in our analysis is presented in Table 1. We divide them into two categories, continuous predictors and

Table 1: Characteristics of cryptocurrencies used as predictors of becoming a zombie

Continuous predictors			
based on	estimation window	statistics	specification
daily returns	28 days	mean, median, min, max, standard deviation, skewness, kurtosis, Q1 and Q3	S1, S2, S3
daily returns	182 days	as above	S1, S2, S3
daily volumes	28 days	as above	S1, S2, S3
daily volumes	182 days	as above	S1, S2, S3
daily market cap	28 days	as above	S1, S2, S3
daily market cap	182 days	as above	S1, S2, S3
daily returns	182 days	Hurst exponent, alpha stable tail index, expected shortfall, value-at-risk	S1, S2, S3
age of a crypto		number of days since the first quotation	S2, S3
Categorical predictors			
variable	values		specification
category	coin/token		S2, S3
has Twitter account	0/1		S2, S3
has subreddit account	0/1		S2, S3
is mineable	0/1		S3
proof-of-stake	0/1		S3
proof-of-work	0/1		S3

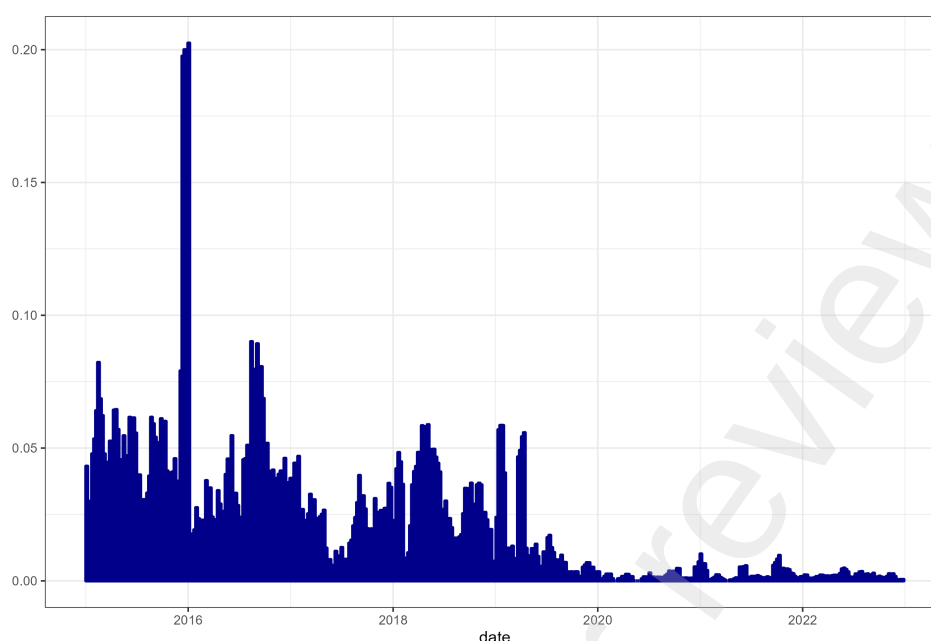
categorical ones. We end up with a set of 58 predictors for Specification 1, 62 for Specification 2, and 65 for Specification 3. While some predictors may exhibit high mutual correlation, this issue is addressed by the regularization features inherent in the machine learning algorithms we use in the empirical part.

4 Empirical results

In this section, we present the empirical results of our analysis using the rolling window approach to train and validate the models. Before we delve into these results, it is important to note that our dependent variable distribution is highly imbalanced. The share of cryptos becoming untradable within the next 28 days by week is illustrated in Figure 4. At the beginning of the analyzed period, around 5% of cryptos were becoming zombies. By mid-2019 the share of untradable symbols decreased to ca. 2.5%. Starting from the second half of 2019 the share of zombies dropped below 0.5%. Due to the highly imbalanced distribution of the dependent variable, we apply three alternative approaches to mitigate the problem and rebalance the training data. These are: simple downsampling (in short DOWN), Synthetic Minority Over-Sampling Technique (SMOTE) (Chawla et al., 2002), and Random Over-Sampling Examples (ROSE) (Menardi and Torelli, 2012).

Table 2 results are obtained from the rolling validation samples approach. They show the performance measures averaged over seven periods. We find that the simplest specification S1, including predictors based only on the statistical properties of returns, volumes,

Figure 4: The share of cryptos becoming zombies in the next 28 days



Note: This plot shows the weekly proportion of cryptocurrencies that lost market quotation coverage relative to the total number of tracked cryptocurrencies in our sample. The sample period spans from January 1, 2015, to December 31, 2022.

and market capitalization, performs as well as or better than more complex models using additional categorical features and the age of the cryptocurrency (S2 and S3).

For comparison, we show the results obtained for predicting whether a symbol will become untradable during the next 7 days. Table 4 shows the performance of all algorithms in the 7 averaged validation samples. Overall, the results do not differ much from the results of predictions for 28 days ahead. The values of all metrics are slightly higher for the 7-day horizon compared to the 28-day horizon. This aligns with intuition, as predicting whether a cryptocurrency will become a zombie over a shorter timeframe (next week) is generally easier than forecasting over a longer period (next month). Simpler specifications with fewer predictors also seem to have higher predictive performance than more complex models that include more information about the cryptocurrencies.

Again, the performance of tree-based models, particularly the random forest trained on a ROSE rebalanced sample, is much higher. This suggests that they better identify the shape of relationships than the traditional fixed-specification models. Tree-based models are adept at capturing complex, non-linear interactions between variables, which likely accounts for their superior performance in these predictive tasks.

In terms of the AUROC statistic random forest seems to slightly outperform the other approaches for all three specifications, in particular for SMOTE and DOWN. However, models trained on simply downsampled training data are based on limited information (a smaller sample) compared to the remaining approaches. Random forest trained with SMOTE rebalancing has in turn the lowest balanced accuracy and sensitivity out of all compared alternatives. Considering these two metrics, and sensitivity in particular, it seems that a random forest trained with ROSE rebalancing is the best-performing approach, followed by the xgboost (also with ROSE).

The above results indicate that the tree-based machine learning models, which catch potential non-linearities and interactions between predictors, perform better than tra-

Table 2: Performance of models predicting becoming a zombie within the next 28 days in the 7 averaged validation samples

rebal.	algo.	AUROC			Balanced accuracy			Sensitivity		
		S1	S2	S3	S1	S2	S3	S1	S2	S3
DOWN	GLM	0.849	0.844	0.844	0.772	0.744	0.767	0.711	0.655	0.713
	LASSO	0.858	0.858	0.857	0.814	0.814	0.801	0.835	0.830	0.806
	RF	0.872	0.870	0.865	0.801	0.805	0.801	0.766	0.789	0.790
	XGB	0.846	0.851	0.849	0.778	0.787	0.790	0.736	0.752	0.763
ROSE	GLM	0.851	0.851	0.847	0.795	0.788	0.784	0.785	0.772	0.763
	LASSO	0.851	0.850	0.842	0.807	0.805	0.797	0.814	0.810	0.802
	RF	0.839	0.832	0.835	0.806	0.797	0.794	0.891	0.861	0.859
	XGB	0.851	0.853	0.851	0.803	0.807	0.784	0.872	0.849	0.820
SMOTE	GLM	0.843	0.846	0.844	0.759	0.737	0.751	0.650	0.601	0.639
	LASSO	0.855	0.853	0.857	0.806	0.794	0.786	0.791	0.763	0.739
	RF	0.878	0.875	0.867	0.729	0.751	0.727	0.534	0.588	0.551
	XGB	0.844	0.850	0.839	0.778	0.771	0.750	0.706	0.688	0.651

Note: The table reports the performance measures of several machine learning models in the prediction of a cryptocurrency becoming a zombie in the 7 averaged validation samples based on 3 alternative specifications, S1, S2 and S2, differentiated by the lists of predictors and three alternative approaches to rebalancing (rebal.). Among the machine learning algorithms (algo.) considered in the study *GLM* stands for the standard logistic regression, *LASSO* is for the least absolute shrinkage and selection operator, *RF* is for the random forest, while *XGB* is for the extreme gradient boosting. Both the performance measures and specifications are described in Section 2. The darker the colour of the cell, the better the model from the point of view of a given performance measure.

Table 3: Performance of models predicting becoming a zombie within the next 28 days in the test sample. The ROSE rebalancing only

algorithm	AUROC			Balanced accuracy			Sensitivity		
	S1	S2	S3	S1	S2	S3	S1	S2	S3
GLM	0.945	0.941	0.932	0.847	0.846	0.847	1.000	1.000	1.000
LASSO	0.937	0.933	0.897	0.840	0.840	0.834	1.000	1.000	1.000
RF	0.897	0.871	0.882	0.848	0.849	0.845	1.000	1.000	1.000
XGB	0.898	0.888	0.886	0.741	0.760	0.788	0.750	0.750	0.750

Note: The table reports the performance measures of several machine learning models in the prediction of a cryptocurrency becoming a zombie in the test sample. The sample covers the second half of 2022. Among the machine learning algorithms considered in the study *GLM* stands for the standard logistic regression, *LASSO* is for the least absolute shrinkage and selection operator, *RF* is for the random forest, while *XGB* is for the extreme gradient boosting. Both the performance measures and specifications are described in Section 2. The darker the colour of the cell, the better the model from the point of view of a given performance measure.

ditional econometric models (GLM) and their extensions (LASSO). We select a random forest trained with ROSE rebalancing as the single best approach consistently for all three specifications and use it later for reasoning about feature importance and the shape of relationships between predictors and the probability of becoming a zombie.

We compare the performance metrics of all applied algorithms and specifications for the ROSE approach on the test sample (Table 3). These metrics confirm that the single best random forest still performs very well with a balanced accuracy of 84%. Although its AUROC is lower compared to GLM or LASSO, it is still very high (88-90% depending on the specification). We can also see out-of-time sensitivity equal to 100%. This is related to extremely rare real cases of cryptos becoming untradable in the test sample (only 16 cases out of ca 59k observations), all of which were predicted correctly by most models.

For comparison, we show the results obtained for predicting whether a symbol will become untradable during the next 7 days. Table 4 shows the performance of all algorithms in the 7 averaged validation samples. Overall, the results do not differ much from the results of predictions for 28 days ahead. The values of all metrics are slightly higher than for 28 days, which, in line with intuition, indicates that predicting the default of a cryptocurrency a week before is easier than 28 days in advance. Simpler specifications with fewer predictors also seem to have higher predictive performance than more complex models that include more information about the cryptocurrencies.

Again, the performance of tree-based models, particularly the random forest trained on a ROSE rebalanced sample, is much higher. This suggests that they better identify the shape of relationships than the traditional fixed-specification models. Tree-based models are adept at capturing complex, non-linear interactions between variables, which likely accounts for their superior performance in these predictive tasks.

To better understand the results, we analyze variable importance based on the previously mentioned best-performing model – random forest, trained on a ROSE rebalanced sample. We use permutation-based feature importance. Figure 5 shows the top 10 predictors for each specification. The top two predictors are the same, regardless of the specification: the minimum (log) volume during the last 182 days (approximately half a year) and the minimum (log) volume in the last month (28 days). The importance of these two features is much higher than that of the remaining ones. This is an important indication for investors – the inactivity in cryptocurrency trading is often preceded by a decline in trading volume. This is justified in the financial literature – volumes are often used as proxies for liquidity (Będowska-Sójka, 2019; Domowitz et al., 2001), and

Table 4: Performance of models predicting becoming a zombie within the next 7 days in the 7 validation samples (averaged)

rebal.	algo.	AUROC			Balanced accuracy			Sensitivity		
		S1	S2	S3	S1	S2	S3	S1	S2	S3
DOWN	GLM	0.860	0.860	0.799	0.769	0.778	0.740	0.710	0.743	0.673
	LASSO	0.889	0.888	0.879	0.871	0.870	0.839	0.936	0.927	0.899
	RF	0.901	0.899	0.894	0.854	0.850	0.844	0.903	0.900	0.865
	XGB	0.618	0.618	0.617	0.616	0.616	0.612	0.273	0.273	0.263
ROSE	GLM	0.881	0.881	0.876	0.796	0.825	0.792	0.774	0.827	0.761
	LASSO	0.875	0.875	0.879	0.845	0.832	0.839	0.900	0.876	0.908
	RF	0.862	0.863	0.858	0.829	0.830	0.828	0.913	0.919	0.921
	XGB	0.882	0.884	0.881	0.843	0.847	0.817	0.919	0.922	0.871
SMOTE	GLM	0.851	0.872	0.865	0.799	0.772	0.786	0.719	0.659	0.697
	LASSO	0.880	0.878	0.880	0.846	0.847	0.820	0.870	0.863	0.814
	RF	0.894	0.888	0.893	0.818	0.781	0.766	0.769	0.685	0.618
	XGB	0.798	0.815	0.788	0.752	0.753	0.745	0.607	0.608	0.588

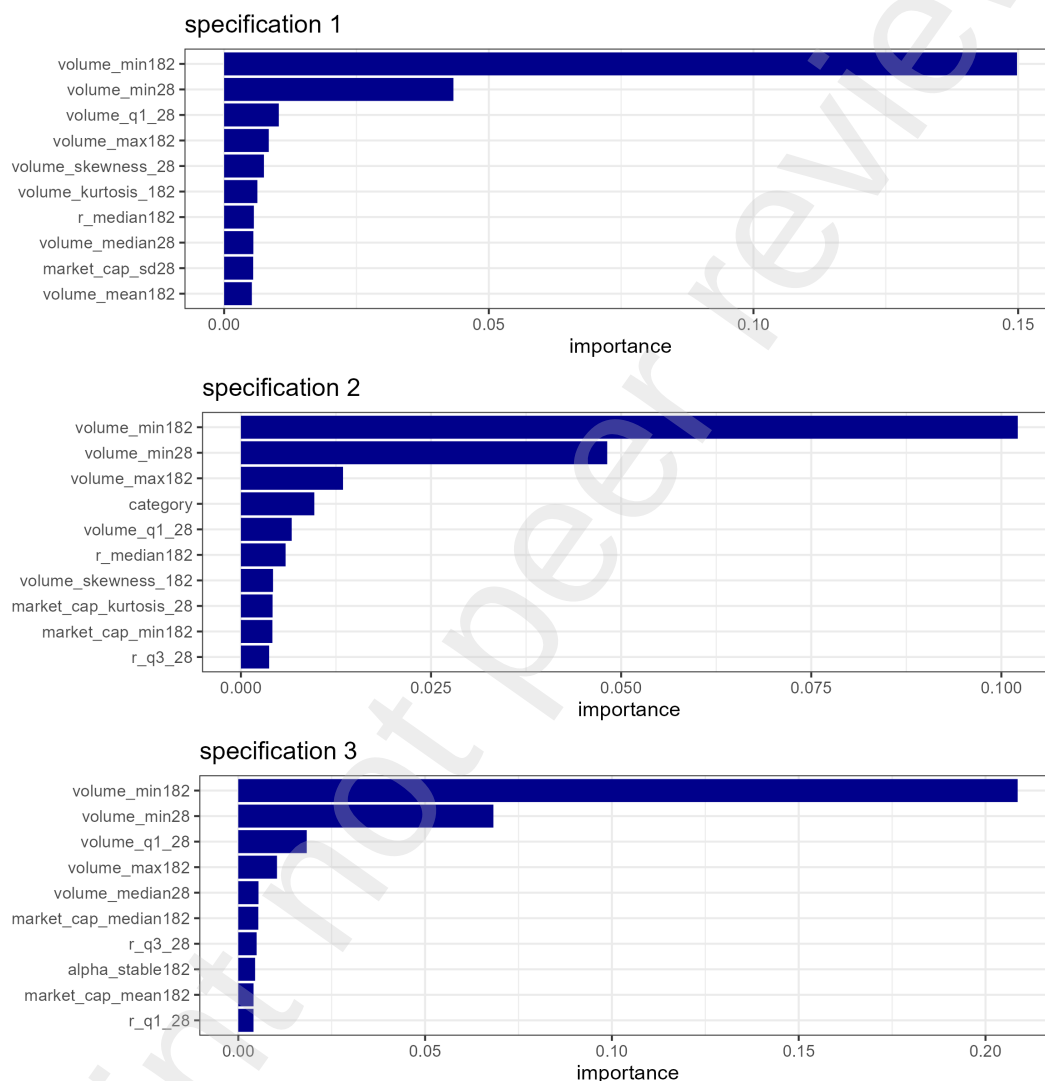
Note: The table reports the performance measures of several machine learning models in the prediction of a cryptocurrency becoming a zombie in the 7 averaged validation samples based on 3 alternative specifications, S1, S2 and S3, differentiated by the lists of predictors and three alternative approaches to rebalancing (rebal.). Among the machine learning algorithms (algo.) considered in the study *GLM* stands for the standard logistic regression, *LASSO* is for the least absolute shrinkage and selection operator, *RF* is for the random forest, while *XGB* is for the extreme gradient boosting. Both the performance measures and specifications are described in Section 2. The darker the colour of the cell, the better the model from the point of view of a given performance measure.

the decrease in liquidity is usually a bad signal for investors. The maximum (log) volume from the last 182 days ranks as the third or fourth most important predictor. Among the predictors based on returns, the median return from the last 182 days appears to be one of the most important features in the first two specifications. The only variable not based on the statistical properties of returns, volumes, or market caps that appear among the top predictors is the category (in the second specification).

Knowing which features influence the probability of becoming untradable the most, we want to identify the shape of their relationships with the outcome of the model. To assess this, we calculated their marginal effects (*ceteris paribus*) and presented them as Partial Dependency Profiles (PDPs) in Figure 6. The analysis is based on relationships observed in the final test sample (2022H2) within the first specification, S1, and the final best random forest model. For comparison, we also show PDPs based on the LASSO model trained with the ROSE rebalancing.

Figure 6 shows that for the minimum logarithms of volumes, both in the last 28 days (*log_volume_min28*) and in the last 182 days (*log_volume_min182*), the relationship is decreasing and monotonic. The larger the minimum trading volume, the lower the risk that a cryptocurrency will become a zombie soon. However, in contrast to the LASSO approach, which suggests an almost constant slope and thus a linear shape, a random forest shows a clear threshold for the 6-month minimum or several thresholds for the minimum of the last 28 days. If the minimum trading volume from the last 6 months is above approximately 2000 (log volume above 7.5), the risk of becoming a zombie is at a stable and very low level. For lower values, it sharply increases, reaching a plateau of approximately 38% for a minimum volume below 15 (log volume lower than 2.5). In the case of the last month (28 days), the probability of becoming a zombie is extremely low

Figure 5: Permuted feature importance for top 10 predictors in specification 1, 2, and 3.



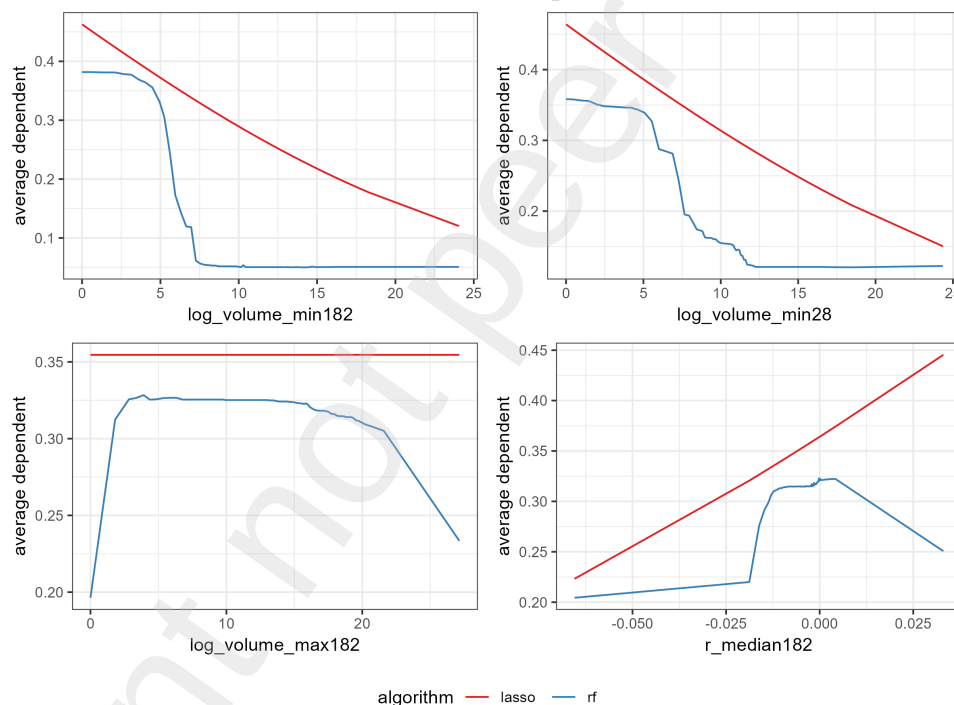
Note: The importance is based on the random forest trained with ROSE rebalancing and applied to the test data. The first part of the variable name refers to a quantity, such as volume, market cap, or returns (r), the second to a descriptive statistic, such as minimum, maximum, mean, median, Q1 or Q3, and the last to the number of days for which the statistic is determined. The only exception is for alpha stable, where there are no descriptive statistics included in the name. Among the variables, volume_min_182 is used for the minimum volume calculated within 182 days. volume_min_28 stands for the minimum volume within 28 days, volume_q1_28 is for the first quantile of the volume distribution, etc.

only for symbols with a minimum volume above 150k (log volume higher than 12).

The relationship between the maximum logarithm of volume from the last 182 days ($\log_volume_max182$) and the outcome probability has a non-monotonic shape based on a random forest. In contrast, LASSO shows a lack of influence of this variable on the risk of becoming a zombie, represented by a flat line. This illustrates the advantage of using non-linear tree-based models. The random forest confirms that the risk of becoming untradable does not depend on the maximum volume for a wide range of its values. However, a cryptocurrency with an extremely large maximum volume in the last 6 months seems to be less risky. Additionally, the risk appears to decrease for very low maximum volumes, which is less intuitive and might require further investigation.

Last but not least, the relationship between the median of returns during the last 182 days ($r_median182$) and the risk of becoming a zombie also shows a non-monotonic shape, resembling an inverted parabola with a peak around zero. This result suggests that symbols whose prices do not move much for an extended period or fluctuate around the same level for 6 months are potentially more risky in terms of becoming untradable.

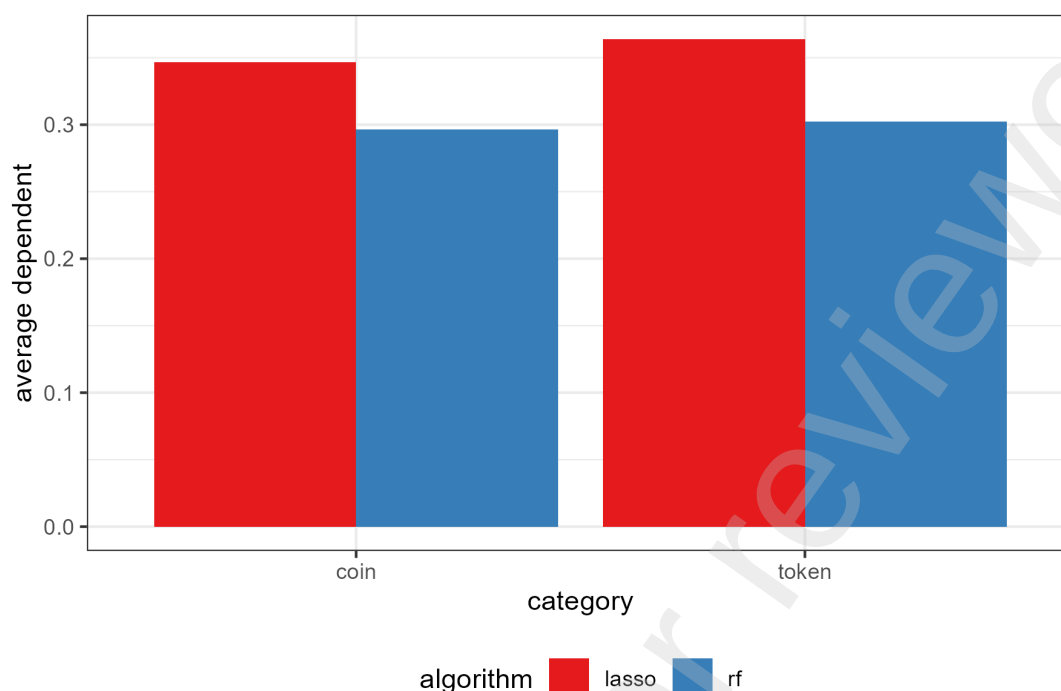
Figure 6: Partial Dependency Profiles for four most important predictors based on S1 trained with ROSE – the results for the test sample



Note: PDPs for random forest and LASSO for comparison, both trained with ROSE rebalancing and applied on the test data. $\log_volume_min182$ stands for the minimum log volume from the last 182 days, $\log_volume_min28$ is for the last 28 days, $\log_volume_max182$ is the maximum of log volumes from the last 182 days, while $r_median182$ stands for median returns based on recent 182 days.

In Figure 7 we present also the Partial Dependency Profile for the category of the crypto, which was slightly important in our second specification. However, based on the random forest the difference in predicted probability of becoming a zombie (having all other characteristics the same) is only marginally higher for the tokens as compared with the coins. Overall, the analysis shows that all of the identified features can be useful indicators of potential risk related to a particular cryptocurrency.

Figure 7: Partial Dependency Profiles for category coin or token based on S1 trained with ROSE – the results for the test sample



Note: PDPs for random forest and LASSO for comparison, both trained with ROSE rebalancing and applied on the test data.

5 Conclusion

In this study, we have conducted an empirical exercise using several machine-learning approaches intended to forecast the probability of a cryptocurrency becoming a zombie within the next period of 7 or 28 days. We deal with risks that investors face due to the disappearance of cryptocurrencies from the market, trying to predict such a situation.

Our results support both hypotheses presented in the Introduction. We were able to predict if a cryptocurrency would become a zombie within the next 28 days with a relatively high balanced accuracy rate of 84%. We found that tree-based algorithms, which are capable of identifying complex, potentially non-linear, and even non-monotonic relationships, are better choices for detecting zombies. The best-performing algorithm is a random forest, with top variables including minimum volume during the last 182 days and 28 days, maximum volume in the last 182 days, and median return in 182 days. We also discovered that simpler specifications, based purely on the statistical characteristics of prices and volumes, have comparable or even better predictive performance than models incorporating the age of the cryptocurrency or categorical predictors, such as the category of the symbol, having an X (former Twitter) account, being mineable, or being based on POS or POW. The only additional feature identified as slightly important was whether the cryptocurrency is a coin or token.

From the investors' point of view, our findings are particularly significant. The ability to predict whether a cryptocurrency is likely to become a zombie within the next 28 days provides investors with critical information to make more informed decisions. Investors can use the insights from our model to identify high-risk assets and avoid potential losses. The variables used in our model, such as trading volumes and returns, are easy for investors to calculate and monitor, making the model practical and accessible for individual and institutional investors alike.

For market regulators, our model offers valuable insights into the risk factors associated with cryptocurrency investments. By understanding the predictors of a cryptocurrency becoming a zombie, regulators can develop better policies to protect market participants. The high accuracy of our model in forecasting zombie cryptocurrencies can help regulators in early detection and intervention, potentially preventing significant market disruptions. Furthermore, as regulations in the cryptocurrency market continue to evolve, our model can serve as a tool for regulators to identify and monitor high-risk assets, thereby enhancing market stability and investor protection.

Since volume and price data are readily available to investors, our results can be used to build early warning systems for the potential to become a zombie. Overall, our model provides a practical and effective means for both investors and regulators to navigate the volatile cryptocurrency market, mitigate risks, and make more informed decisions based on robust empirical evidence.

In this study, we focus on predicting the likelihood of cryptocurrencies becoming zombies, defined as assets that either disappear from the market or become untradable. While our approach provides valuable insights, certain limitations should be acknowledged. We rely on data from CoinMarketCap, a widely recognized and frequently used source in cryptocurrency research. However, its evolving criteria for including assets may introduce some variability into our sample.

Our analysis primarily utilizes financial metrics, such as trading volume and returns, as predictors. While these factors are central to understanding zombification, incorporating additional qualitative and fundamental variables could enhance the depth of the analysis. Additionally, examining exchange-level details, such as distinctions between centralized and decentralized platforms, could provide more nuanced insights into the dynamics that lead to zombification. In future work, we plan to address these limitations by exploring longer forecast horizons, expanding the set of predictors, and incorporating more granular data on trading activity to improve the understanding of this phenomenon.

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